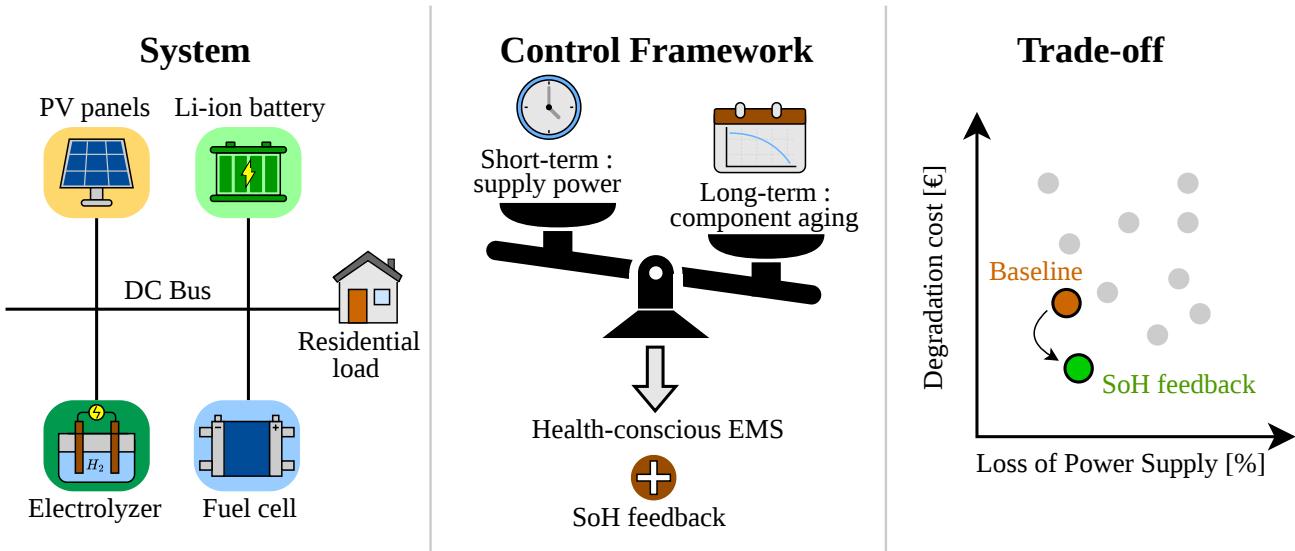


Graphical Abstract

A Methodological Framework for Health-conscious Energy Management in Microgrids: Balancing Service Availability and Component Aging



Highlights

A Methodological Framework for Health-conscious Energy Management in Microgrids: Balancing Service Availability and Component Aging

- A framework jointly optimizes power supply reliability and storage component aging
- State-of-health monitoring enables adaptive energy management in hybrid microgrids
- Aging feedback in rule-based control reduces replacement costs by about 10% at constant service quality
- Start-stop cycles are the dominant degradation factor for fuel cells and electrolyzers

A Methodological Framework for Health-conscious Energy Management in Microgrids: Balancing Service Availability and Component Aging

Abstract

This paper presents a methodological framework for energy management in hybrid microgrids integrating photovoltaic panels, batteries, and hydrogen electrolyzers and fuel cells. The proposed approach addresses the dual objective of ensuring power availability for end-users in the short-term while preserving the long-term sustainability of energy storage components. A macroscopic model of the system, including performance indicators and state-of-health estimation methods for the storage devices, is provided. This model is used for the analysis of several energy management strategies, which are compared using performance indicators quantifying both power supply reliability and replacement costs induced by degradation. The most promising rule-based strategy is then enhanced by introducing an adaptive layer that updates control rules according to the estimated state-of-health of the storage components. Simulation results demonstrate that this adaptive approach can reduce component degradation, and the associated replacement costs, without compromising service continuity. The study highlights the relevance of explicitly integrating component aging into energy management strategies, reducing replacement costs by about 10% while keeping the loss of power supply probability essentially unchanged.

Keywords: Hybrid microgrid, Energy management strategy, Component aging, Rule-based control, State-of-health, Energy storage systems

1. Introduction

1.1. Context and problem statement

Hybrid electric systems—i.e. power systems that combine multiple energy storage technologies—are widely represented in power engineering. They allow leveraging intermittent renewable generation (solar, wind, etc.) alongside storage units and auxiliary generators (fuel cells, electrolyzers, etc.) to ensure reliable power supply, even in isolated settings.

In this article, a hybrid DC microgrid in an isolated operation context located on La Réunion Island (France) is studied. It supplies electricity to residential users via photovoltaic (PV) arrays, combined with a hybrid storage/conversion system composed of lithium-ion batteries, a PEM (Proton Exchange Membrane) electrolyzer producing hydrogen, and a PEM fuel cell (Fig. 1). The PV panels are the only energy source, and the storage systems act as buffer layers for times when consumption is not aligned with production, which is intermittent by nature. The battery system's faster dynamics allow for quick buffering to smooth short-term fluctuations, while the hydrogen chain allows longer duration balancing. Such microgrids are broadly used, both in literature [1][2] and real-world applications [3][4].

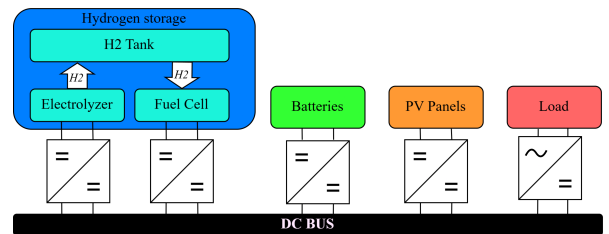


Figure 1: General architecture of the hybrid microgrid

Because of this multi-storage / multi-conversion structure, there are many degrees of freedom in the power sharing: a controller must decide in real time how much power from PV to use directly to supply loads, how much to charge/discharge the battery, how much to send to the electrolyzer, or draw from the fuel cell, etc. These control decisions are also taken under given constraints (battery state-of-charge, hydrogen availability, maximum power ratings of the components), and are taken at hourly intervals for a rather macroscopic approach.

Energy management strategies (EMS) significantly influence the macroscopic behavior of microgrids in

multiple ways, from costs to reliability, degradation and maintenance requirement. Assessing these strategies through performance indicators often relies on measurements conducted over different time constants, spanning from milliseconds to years, which induces computation time issues. Prioritizing one indicator over another implies different impacts on the microgrid and its components. For instance, focusing on maintaining high battery state-of-charge (SoC) levels might help mitigate short-term power fluctuations. However, this would lead to a more intensive use of hydrogen components and, consequently, greater degradations and aging. Short-term gains would result in long-term losses. To address these conflicting objectives, this work proposes a methodological framework based on multi-timescale indicators. This approach enables a comprehensive evaluation of energy management, facilitating the design of EMS and reaching a robust compromise between short and long-term requirements, while also considering user expectations.

1.2. State-of-the-art

1.2.1. Objectives for EMS

The objectives of the EMS are fundamentally dictated by the time constants of the targeted phenomena, which are often addressed in isolation within the literature. Several studies focus exclusively on short-term dynamics to ensure system stability and power quality. For instance, Hu et al. implement a model-free adaptive control for battery-supercapacitor systems to maintain DC bus voltage stability and track load variations on a scale of seconds to minutes [5]. Similarly, Asrari et

al. develop a programmable controller focusing on real-time solar tracking and power quality, using heuristic current limitations to mitigate aging [6]. While effective for immediate power balancing, these approaches typically overlook long-term degradation mechanisms and operational constraints spanning hours or days.

At an intermediate scale, research shifts toward economic optimization and peak load management. Paulraj et al. utilize LSTM networks to reduce battery peak currents and energy consumption, offering a medium-term perspective that optimizes energy flow over seconds to hours [7]. However, this approach remains limited to a single time constant and does not account for cumulative aging over longer periods. In the context of microgrids, Guo et al. propose a hierarchical economic model predictive control (MPC) to manage hydrogen component start-stop cycles and align operation with weather forecasts and electricity markets [8]. Although this framework bridges the gap between hourly and weekly scales, it does not explicitly integrate component degradation or long-term sustainability beyond the weekly horizon.

A distinct branch of the literature addresses long-term objectives such as seasonal energy shifting and multi-year component health. Qi et al. introduce a two-stage optimization coordinating hourly supply-demand balance with seasonal hydrogen storage to mitigate renewable intermittency over year-long horizons [9]. Despite its long-term scope, this study does not account for the physical deterioration of the storage components. Conversely, Wang et al. focus specifically on reducing electrochemical degradation in fuel cells and batteries over months and years, though their model as-

Table 1: Summary of the state-of-the-art regarding EMS objectives and time constants.

Ref.	Contribution	Research Gap
[5]	Real-time load tracking and DC bus voltage stability (seconds to minutes).	No long-term aging, no dynamics beyond the minute scale.
[6]	Power quality enhancement and solar tracking with heuristic current limits.	Limited to a single short-term time constant, no loop on long-term aging.
[7]	Optimization of battery peak current and energy consumption (seconds to hours).	Lacks long-term health monitoring; it is a unique intermediate timescale
[8]	Hierarchical economic MPC for H2 management and market/weather integration.	Does not explicitly incorporate physical degradation or long-term aging.
[9]	Coordination of seasonal energy shifting and long-term renewable intermittency.	Omits the impact of component deterioration on system performance.
[10]	Focus on multi-year degradation reduction for fuel cells and batteries.	Assumes short-term power fluctuations are pre-satisfied by an external layer.
[11]	Multi-decade evaluation of hybrid systems considering capacity fading.	Aging is treated as an exogenous indicator rather than an active control input.

sumes that the short-term load-following requirements are pre-satisfied, thus neglecting real-time power fluctuations [10].

The challenge of multi-scale integration is partially addressed by Song et al., who evaluate an off-grid system over a 20-year lifetime [11]. Their work combines hourly load tracking with long-term cost reduction and capacity fading analysis. However, in this framework, aging is treated as an exogenous indicator rather than an active input for the control loop, meaning the strategy does not dynamically adapt to the evolving state-of-health (SoH) of the components. This highlights a persistent gap in the literature: the need for a methodological framework that explicitly couples short-term power reliability with an active, health-aware management of energy storage over the long term. This section is summarized using Table 1.

1.2.2. EMS structures

There are then several possibilities to construct EMS that can achieve these objectives, which will be addressed. Published approaches can be broadly grouped into (i) rule-based, (ii) optimization-based, and (iii) data-driven. This part will focus on online EMS only as the end goal is to implement it in the actual microgrid.

Optimization-based strategies often rely on MPC to handle multi-objective constraints. For instance, Kbidi et al. developed a multistage power and energy management strategy coupling distributed explicit MPC with mixed-integer quadratic programming to optimize unit commitment and minimize fuel cell start-stop cycles [12]. Similarly, Hou et al. proposed a hierarchical MPC algorithm that decomposes the problem into a long-term strategic layer for battery SoC tracking and a short-term tactical layer for real-time power distribution [13]. While these methods achieve high theoretical efficiency, their performance is sensitive to prediction accuracy and requires significant computational power for real-time execution.

Data-driven methods, particularly Reinforcement Learning (RL), offer an alternative by learning control policies through interaction with the system. Han et al. demonstrated that a Double Deep Q-Learning strategy could improve fuel economy compared to standard rules by optimizing steady-state processes [14]. To address the stability limits of pure learning approaches, recent studies explore hybrid structures. Ren et al. combined Deep Q-Networks with an Equivalent Consumption Minimization Strategy (ECMS) to adaptively tune the equivalence factor based on driving experience [15]. In a similar effort to balance performance and stability,

Liu et al. integrated MPC with RL, where the former ensures constraint satisfaction while the latter provides a compensatory policy for long-term decision-making [16].

Despite the performance gains reported for predictive and learning-based methods, rule-based strategies remain a standard for industrial applications due to their transparency and low computational cost. Nivolianiti et al. utilized fuzzy logic to distribute power in a hydrogen-powered vessel, highlighting that such rules operate effectively without requiring future information [17]. The trade-offs between these paradigms were analyzed by Wu et al., who compared pure rule-based, optimization-based, and hybrid strategies, concluding that while hybrid models improve efficiency, rule-based foundations offer the highest operational robustness [18].

Building on these observations, this study adopts a rule-based methodological framework. Rather than increasing algorithmic complexity, the proposed approach focuses on a macroscopic system model that explicitly incorporates component aging and SoH estimation. This choice allows for a deterministic and computationally efficient EMS, providing a clear basis for evaluating the trade-offs between short-term power availability and the long-term sustainability of energy storage components.

1.3. Objectives of the study

Considering the works showcased in recent literature, the present work aims to jointly address short-term service performance and long-term system sustainability in the energy management of hybrid microgrids, while actively looping on component aging. Most existing EMS frameworks emphasize either operational efficiency—minimizing instantaneous cost, fuel use, or unmet demand—or component preservation through degradation cost terms. However, few approaches explicitly attempt to consider both dimensions simultaneously, particularly when the underlying phenomena evolve on inherently different timescales: power balance and service continuity operate on the order of seconds to minutes, while component degradation unfolds over months, or years. This will be the scope of this article.

The first objective of this study, namely the short-term objective, is to ensure that user demand is satisfied as much as possible given the available renewable generation and storage capacity. This involves minimizing load shedding, and also eventually power curtailment to ensure high enough levels of hydrogen storage and battery SoC. The short-term strategy thus reflects the

system's operational adequacy—its ability to maintain a stable and continuous supply despite renewable variability and the physical limitations of storage devices.

The second objective, the long-term objective, concerns the durability of the hybrid system, defined here in terms of the aging behavior of its key components. Batteries, electrolyzers, and fuel cells all undergo wear mechanisms linked to usage patterns, SoC cycling, and thermal or electrical stress. Excessive solicitation of one subsystem (e.g., repeated deep cycling of the battery) may improve short-term service but accelerate degradation, reducing the system's overall lifetime and economic viability. The long-term objective therefore introduces cost indicators that quantify degradation over time, allowing the EMS to assess and balance the consequences of control decisions on component health.

In doing so, the present work positions itself as a methodological step toward multi-scale, health-conscious control of hybrid microgrids, in which operational and aging dynamics are treated not as independent concerns but as interdependent aspects of an integrated power management problem. These two objectives may prove contradictory and introduce the need for trade-offs, which will be determined by local contexts.

This next section will describe the case study in more detail, as well as the mathematical formulation of the problem. Performance indicators will be introduced, followed by methods for estimating the SoH of components. Finally, the simulation results will be presented, allowing for an informed comparison of the different strategies studied in this work.

2. Model description

2.1. Timescales and methodological positioning

Microgrid operation inherently spans multiple timescales, ranging from instantaneous power balancing to long-term component degradation (Fig. 2).

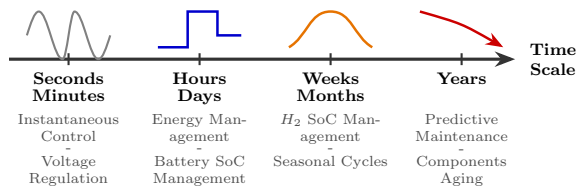


Figure 2: Conceptual overview of the overlapping operational timescales, the hourly resolution, and the multi-year simulation horizon.

The proposed framework explicitly addresses this duality by separating two operational horizons: the short-term horizon, dedicated to localized power dispatching (matching PV generation and load demand while minimizing power losses), and the long-term horizon, which tracks component aging (evaluating battery, fuel cell, and electrolyzer SoH over several years).

To bridge these contrasting scales within a computationally tractable framework, an hourly temporal resolution is adopted. Meteorological inputs, and user demands are assumed constant within each discrete time step. This resolution is fine enough to capture daily cyclic dynamics while remaining sufficiently coarse to allow multi-year simulations (e.g., 5 to 20 years), which are necessary to evaluate total lifecycle and component replacement costs.

This requirement for long-term simulation horizons directly dictates the choice of the control paradigm. While optimization-based strategies (such as MPC) or data-driven approaches are highly effective for short-term dispatch, their computational cost at each time step becomes prohibitive over multi-year horizons. Conversely, the selected rule-based framework evaluates explicit logic instantaneously. This ensures an acceptable computational burden per step, making it suited for the long-term simulations required to analyze health-conscious energy management trade-offs.

2.2. System overview

The system under study is an isolated hybrid DC microgrid designed to supply a group of residential users with electricity on La Réunion island (France), where renewable integration is a necessity due to geographical isolation. The considered architecture combines PV generation, lithium-ion NMC battery storage, and a hydrogen side composed of a PEM electrolyzer (PEMWE) and a PEM fuel cell (PEMFC), see Fig. 1. The system operates around a common DC bus, ensuring electrical interconnection between components and enabling power management.

During daylight hours, the PV generator produces electrical power P_{PV} that can be (i) directly supplied to the load, (ii) stored in the battery, or (iii) converted into hydrogen through the electrolyzer. During low or no PV production periods, the load is supplied by the storage systems. The power split between them is in both cases handled by the EMS.

This study is focused on the energy management of a microgrid considering a given sizing, as it is not in the scope of this work. For this article, the sizing has been roughly tuned based on field observations. Specifically, the PV array is significantly oversized relative

to the demand, leading to a correspondingly oversized electrolyzer compared to the fuel cell. The storage capacity is designed to ensure several days of autonomy. A summary of the main system parameters and nominal ratings is provided in Table 2.

Table 2: Main parameters and sizing of the hybrid microgrid.

Description	Symbol	Nominal value
Average load power	$\bar{P}_{\text{Load}}$	2.2 kW
Maximum load power	$P_{\text{Load,max}}$	4.2 kW
Maximum PV power	$P_{\text{PV,max}}$	32.7 kW
Nominal battery capacity	$E_{\text{bat,nom}}$	39 kWh
Nominal PEMFC power	$P_{\text{FC,max}}$	1.5 kW
Nominal PEMWE power	$P_{\text{ELY,max}}$	16 kW
Hydrogen storage capacity	$E_{\text{H2,max}}$	200 kWh

The battery SoC and hydrogen energy stored E_{H2} [J] represent the state of the system, and their evolutions between time t and $t + \Delta t$ are dictated by:

$$\text{SoC}(t + \Delta t) = \text{SoC}(t) - \frac{P_{\text{bat}}(t) \Delta t}{E_{\text{bat,nom}}} \cdot \eta_{\text{bat}}^k \quad (1)$$

$$E_{\text{H2}}(t + \Delta t) = E_{\text{H2}}(t) - (P_{\text{ELY}}(t) \eta_{\text{ELY}}(t) + \frac{P_{\text{FC}}(t)}{\eta_{\text{FC}}(t)}) \Delta t \quad (2)$$

where P_{bat} [W] is the battery power, considered positive when the battery is discharging, $E_{\text{bat,nom}}$ [J] is the nominal energy stored in the battery, and η_{bat}^k is the battery efficiency, with k being equal to 1 when the battery is charging, and -1 when it is discharging. The efficiency is here considered constant, equal to 0.95. For the hydrogen storage, P_{ELY} [W] is the PEMWE power, negative by nature as it can only draw power, P_{FC} [W] is the PEMFC power, positive by nature as it can only produce power, η_{ELY} and η_{FC} are the efficiencies of the PEMWE and PEMFC, respectively, which depend on the operating point [19][20]. The DC bus power balance then ensures conservation of power at every time step:

$$P_{\text{Load}} - P_{\text{PV}} = P_{\text{FC}} + P_{\text{ELY}} + P_{\text{bat}} \quad (3)$$

where P_{Load} [W] is the power consumed by the users, considered positive, P_{PV} [W] is the power provided by the PV panels, considered positive. These two values can be different from the actual $\hat{P}_{\text{Load}}$ and $\hat{P}_{\text{PV}}$, representing the planned power trajectory derived from the load profiles. This difference may arise, for example, if demand cannot be met by the components or if curtailment is implemented.

In the proposed formulation, the control variables of the system are the battery power P_{bat} , the fuel cell

power P_{FC} , and the electrolyzer power P_{ELY} . This establishes the optimization as a three-dimensional control problem. Once P_{bat} , P_{FC} , and P_{ELY} are determined, the overall system power balance is assessed, eventually being different from the reference trajectory. This flexibility allows the system to actively not follow the power demand at some points if the reduction of systems degradation is considered more important. Operational constraints, such as maximum power limits and hydrogen storage capacity, are respected during this allocation. The power profiles used in this work come from local data given by the designer of this microgrid in La Réunion. Field surveys were conducted on the consumption habits of local residents. This enabled the creation of typical consumption profiles for each family, which were then aggregated and can be seen in Fig. 3.

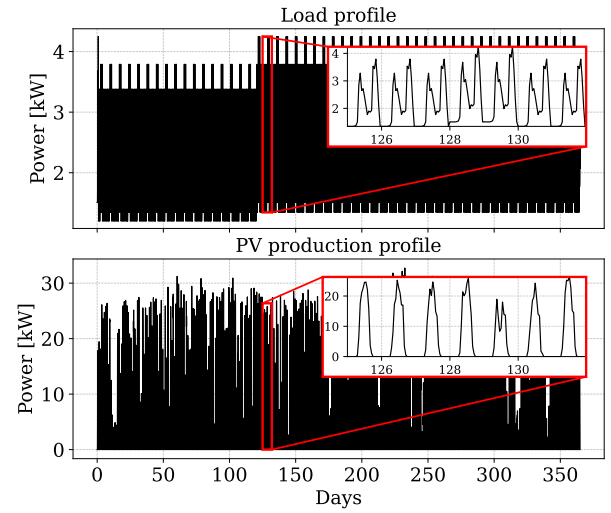


Figure 3: Power profiles used in simulation: load demand and PV generation

2.3. Key indicators definitions

In order to evaluate the performance of the proposed energy management strategies, two complementary indicators are introduced: one reflecting the short-term operational adequacy, and another representing the long-term sustainability, referred to as the degradation indicator. These two metrics respectively quantify the ability of the system to continuously satisfy power demand, and the rate at which the main storage components deteriorate due to operational stress.

2.3.1. Short-term performance: definition of Loss of Power Supply Probability

In the context of this study, the quality of service is assessed strictly from the perspective of the consumer's

continuity of supply. Consequently, the focus is placed on the power deficit. Situations of energy surplus, where the renewable generation exceeds both the load demand and the charging capacity of the ESS, also known as curtailing, are not directly penalized in this specific metric. It is assumed that during such intervals, the control strategy adapts the Maximum Power Point Tracking (MPPT) algorithm to curtail the solar production, thereby maintaining the stability of the DC bus without affecting the service delivery to the load.

Therefore, the primary indicator selected to quantify reliability is the Loss of Power Supply Probability (LPSP). This metric is widely adopted in the energy management and sizing of hybrid renewable energy systems as a standard benchmark [21][22].

The LPSP is defined as the fraction of the total load demand that cannot be satisfied by the hybrid system over a specific time horizon. It ranges from 0 to 1, where 0 implies that the load is fully satisfied at all times, and 1 implies that the load is never satisfied. Mathematically, the LPSP can be expressed as the ratio of the accumulated energy deficit to the total energy demand over the whole simulation period, as shown in Eq. 4:

$$LPSP = \frac{\sum_t LPS(t) \cdot \Delta t}{\sum_t \hat{P}_{Load}(t) \cdot \Delta t} \quad (4)$$

where $LPS(t)$ [W] represents the Loss of Power Supply at time t , defined as:

$$LPS(t) = \begin{cases} \hat{P}_{Load}(t) - P_{Avail}(t) & \text{if } \hat{P}_{Load}(t) > P_{Avail}(t) \\ 0 & \text{otherwise} \end{cases} \quad (5)$$

where $\hat{P}_{Load}(t)$ [W] is the power demanded by the load at time step t , $P_{Avail}(t)$ [W] is the total power available from the PV generation and storage systems at time step t .

2.3.2. Long-term performance: definition of degradation indicator

While LPSP captures short-term adequacy, long-term performance is associated with the rate of degradation of the system's main storage and conversion components. The degradation indicator D_{tot} is defined as the sum of three normalized sub-indicators corresponding to the battery, the PEMFC, and the PEMWE:

$$D_{tot} = C_{bat} D_{bat}^* + C_{FC} D_{FC}^* + C_{ELY} D_{ELY}^* \quad (6)$$

where D^* terms represent normalized degradation indices (dimensionless). The normalization is applied

considering the end-of-life (EoL) criteria of each component, so that each D^* takes values between 0 and 1. The EoL criteria are respectively 40% loss of capacity for the batteries, and 10% increase/loss of power for a constant current for the PEMWE/PEMFC. The C terms correspond to weighting coefficients proportional to the component replacement costs. They are calculated using 90% of the CAPEX for the batteries as well as 90% of the CAPEX of the stack for the PEMWE and PEMFC [23][24][25].

Battery degradation indicator. The indicator of the degradation of the lithium-ion battery, D_{bat} , is constructed to be computed as a function of several stress factors commonly cited in the literature: SoC operating level, depth-of-discharge (DoD), and charge/discharge rate (C-rate) [26][27].

The indicator used in this work is based on a study from Maheshwari et al. [28], which incorporates these three aging factors and it only considers the capacity loss of the battery and ignores, for instance, the increase of the internal resistance. The aging process is firstly described through a graph shown in Fig. 4.

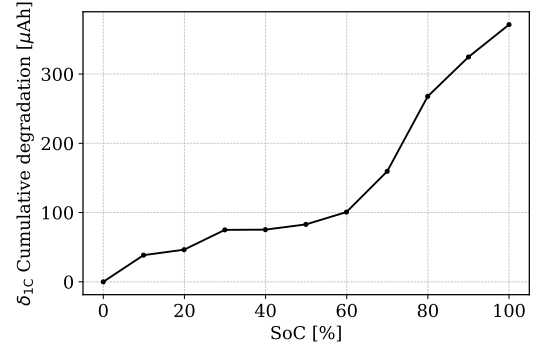


Figure 4: Cycling induced battery capacity degradation [28]

Fig. 4 describes the degradation induced between each timestep where the SoC evolves. The degradation is expressed in μAh , representing the capacity loss in the battery. It also depends on the operating conditions, being the depth of discharge and the SoC at which the battery is used. These degradations δ_{1C} are here presented for a 1C current, and another multiplying factor ψ_t takes the C-rate into account to obtain the real degradation [28]:

$$\delta_t = \delta_t^{1C} \times \psi_t(C\text{-rate}) \quad (7)$$

The total battery degradation over the whole profile is then expressed as:

$$D_{bat} = \sum_t \delta_t \quad (8)$$

Only cycling-induced aging is considered, while calendar aging is neglected, as the proposed indicator is specifically designed to reflect degradation mechanisms directly influenced by energy management decisions.

PEMFC degradation indicator. For the fuel cell, degradation D_{FC} depends primarily on dynamic operation patterns, namely start-stop events, idling periods, load transients, and high-power operation [29][30]. The degradation indicator constructed in this work will therefore depend on these four mechanisms (Eq. 9).

$$D_{FC} = d_{\text{start-stop}} + d_{\text{idling}} + d_{\text{transient}} + d_{\text{high power}} \quad (9)$$

All these factors are defined through a study based on experimental data from Song et al. [31], and are expressed as percent of voltage drop. Here the different factors are obtained through the historical timeserie of the fuel cell power, and $d_{\text{start-stop}}$ associates a voltage drop to the amount of start-stop cycles, d_{idling} associates it to the time spent under a power threshold (1% of the nominal power), $d_{\text{transient}}$ associates it to the relative change in operating power, and finally $d_{\text{high power}}$ associates it to the time spent over a power threshold (80% of the nominal power).

PEMWE degradation indicator. The electrolyzer degradation D_{ELY} relies on a reversible/irreversible voltage-increase model, calibrated on the durability tests reported by Rakousky et al. [32]. Unlike a purely cumulative description, this model captures the fact that part of the PEMWE performance loss is recoverable: a permanent (irreversible) contribution, associated with mechanisms such as titanium porous-transport-layer corrosion and ohmic resistance growth, coexists with a reversible contribution that builds up during operation and partially recovers when the current is interrupted. This model also includes typical degradation mechanisms for electrolyzers, such as idling operation or start-stop events [33]. The total indicator is expressed as the sum of four terms:

$$D_{ELY} = D_{ELY}^{\text{irr}} + D_{ELY}^{\text{rev}} + d_{\text{start-stop}} + d_{\text{idling}} \quad (10)$$

all expressed as a percentage increase in cell voltage. The irreversible and reversible parts follow $\dot{V}_{\text{irr}} = a(i)$ and $\dot{V}_{\text{rev}} = b(i) - k(i) V_{\text{rev}}$, where the generation rates a and b vanish below 1 A/cm² ($\approx 30\%$ of P_{max}), increase linearly up to their rated values at 2 A/cm² ($\approx 60\%$), and saturate beyond. The recovery rate $k(i)$ is large at near-zero current—yielding an almost instantaneous reset at shutdown—and negligible in operation, so that

sustained high-power use degrades the stack almost as a constant-current stress. Finally, $d_{\text{start-stop}}$ adds a fixed voltage increase at each OFF $\rightarrow$ ON transition, while d_{idling} penalizes the time spent at very low power, with a rate taken from Lu et al. [19].

Normalization and aggregation of the indicators. Each of the three degradation indicators is then normalized with respect to their respective end-of-life criteria (see Section 2.4), allowing for direct comparison across technologies. They are then also weighted with the financial replacement cost of the component [25][24]. The total degradation index D_{tot} thus represents an aggregated, cost-weighted measure of the long-term sustainability of a given EMS strategy (see Eq. 6).

2.4. State of health definitions and evaluation methods

In this system model, components are subject to aging throughout the simulation. Their health status must therefore be defined. Unlike the previous section, which aimed to develop a performance indicator for EMS, this section will enable the control system to monitor the health status of components. The aim is to adapt the control system by adjusting to new operating ranges (capacity, power, etc). These health statuses are of course closely linked to the degradation indicators defined above, but they do not serve the same purpose. They do not seek to quantify the performance of an EMS, but rather to observe the aging of components over time and to ensure that they continue to operate within their intended operating ranges (modified SoC ranges for batteries, modified permissible power ranges for fuel cells and electrolyzers).

In this work, the SoH of each main component is represented by a single scalar indicator reflecting its performance degradation relative to nominal. Standard approaches to define SoH for these storage components in stationary applications are considered, and thus:

- For the battery, the SoH is the measured residual capacity relative to the nominal capacity [34].
- For the fuel cell, the SoH is the percentage of power loss relative to rated power at the beginning of life, at constant current rate set at 75% of the maximum power current. [25].
- For the electrolyzer, similarly, the SoH is the percentage of electrical power increase relative to rated power at the beginning of life, where a constant current rate of 75% of the maximum power current is also used [25].

which can also be written:

$$\text{SoH}_{\text{bat}} = \frac{C_{\text{bat, meas}}}{C_{\text{bat, nom}}} \quad (11)$$

$$\text{SoH}_{\text{FC}} = \frac{P_{\text{FC}}(i_{\text{FC, nom}}, t)}{P_{\text{FC}}(i_{\text{FC, nom}}, t = 0)} \quad (12)$$

$$\text{SoH}_{\text{ELY}} = \frac{P_{\text{ELY}}(i_{\text{ELY, nom}}, t = 0)}{P_{\text{ELY}}(i_{\text{ELY, nom}}, t)} \quad (13)$$

Having defined these SoH metrics, several methods exist for their evaluation. These approaches will be summarized briefly as their detailed analysis falls outside the scope of this study.

2.4.1. Online estimation of battery SoH via internal resistance

A widely used proxy for battery aging in the literature is also the increase of internal resistance (or internal impedance), which correlates (negatively) with capacity fade [35]. It is then possible to implement in situ measurement of internal resistance (e.g. via short pulses, or impedance spectroscopy) and infer an estimated capacity, using a previously established correlation model. It will then be considered that existing estimation methods allow the user to track the SoH of the battery at the rate of at least once a week. Since the time constants involved in component aging are typically on the order of several months or years, intra-week variations in SoH will be negligible.

2.4.2. Online estimation of PEMFC and PEMWE SoH via extended Kalman observers

For the hydrogen conversion devices, we adopt model-based diagnostic techniques based on extended Kalman filters (EKF) or variants (e.g. unscented Kalman filter) [36][37]. These methods estimate internal degradation states or parameter drifts from observable measures such as voltage, current, and temperature. The method used to estimate the fuel cell SoH is described in our previous work, using an EKF [38]. It is considered that this method, based on a voltage model, can be extended to electrolyzer applications by reversing the model [39].

2.5. Model parameters and assumptions

The simulation parameters are defined to capture the long-term evolution of the hybrid microgrid operation under various energy management strategies. The total simulation horizon is set to twenty-five years. This duration allows to clearly see the aging processes at works in the storage components, including several component

replacements, while the computation time remains reasonable. Simulations were implemented in Python 3.12 and run on an Intel Xeon W-2123 workstation (4 cores, 3.60 GHz, 32 GB RAM). The average computation time per run was about 10 minutes.

The sampling time for the simulation is fixed to one hour. Only the hourly-sampled control is considered in this work, and control for smaller timescales is considered to be covered by a different layer. Power demands within one hour are then considered to be constant. However, it must be acknowledged that this temporal resolution may lead to a partial underestimation of intra-hour power fluctuations and consequently of some degradation phenomena that could occur under faster operating conditions [40][41].

All simulations are performed using the same environmental conditions and component models to ensure the comparability of results. The photovoltaic power and load demand profiles are based on representative data for a residential microgrid located on La Réunion island (Fig. 3). The assumption is also made that the estimation of the SoH of the components is sufficiently accurate to have associated operating ranges that correspond to reality.

2.6. Simulation algorithm

Once all these premises and assumptions have been established, the simulation algorithm will now be described. At each simulation time step t , the energy management strategy determines the control variables of the system, namely the storage components powers $\tilde{P}_{\text{bat}}(t)$, $\tilde{P}_{\text{FC}}(t)$, $\tilde{P}_{\text{ELY}}(t)$. This decision is based on the instantaneous system state, including the battery SoC, the hydrogen level, the estimated SoH of the storage components, and the power demand from the load and generation sources. If those powers are outside their operating ranges, then a discrepancy between the power demand and the actual power may appear. Once the control variable is defined, the dynamic equations governing the behavior of the components are solved over the current time step to obtain the system state at the next time step. The general simulation loop is summarized in Fig. 5. This workflow is applied for the whole simulation, which is done using the main parameters described in Section 2.5, and several times in a row for each EMS that will be considered. At the end of the simulation, each EMS is thus represented by a single point in a two-dimensional performance space, whose axes correspond respectively to the short-term and long-term objectives (LPSP and degradation costs). This graphical representation will then enable a direct comparison of strategies,

highlighting potential trade-offs between service quality and sustainability.

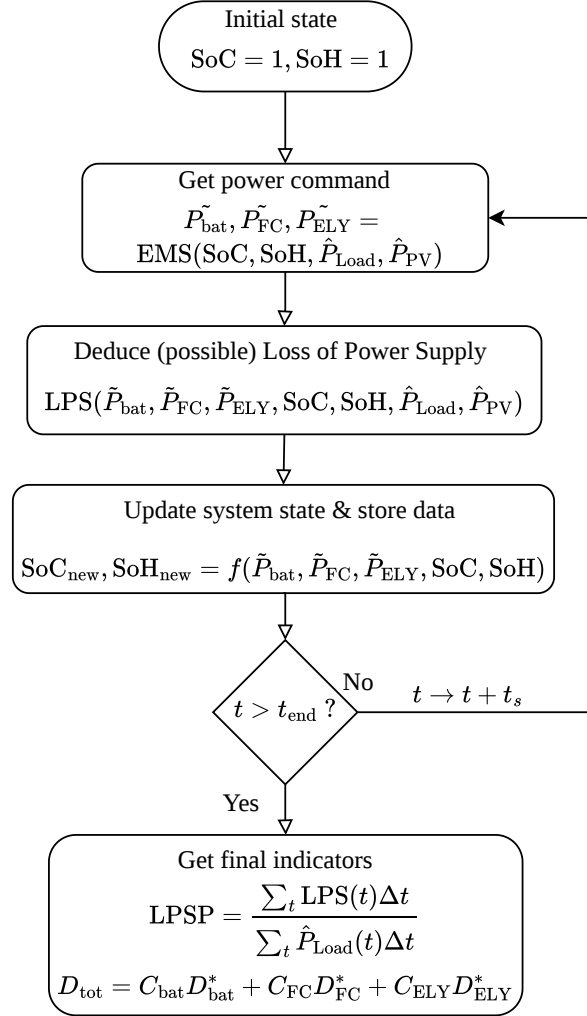


Figure 5: General workflow of the simulation algorithm for evaluating energy management strategies

2.7. Naive EMS comparison

To evaluate the performance of various control logics within the hybrid microgrid, several EMS were designed and simulated. This section will describe the philosophy behind each strategy before comparing them all at once using a Pareto chart.

Power splitting strategies

The first set of strategies consists of static power-sharing rules between the battery and the hydrogen subsystem. These configurations are characterized by fixed

distribution ratios of the total planned power $\hat{P}_{\text{tot,plan}} = (\hat{P}_{\text{Load}} - \hat{P}_{\text{PV}})$ between the two storage media:

$$(100 - 0), (75 - 25), (50 - 50), (25 - 75), (0 - 100)$$

where the first term corresponds to the share [%] attributed to the battery and the second to the hydrogen system. These strategies therefore go from a philosophy attributing most of the work to the battery to attributing it to the hydrogen systems [42]. This approach has a similar structure to the more common frequency splitting strategies, where two storage components having different time dynamics share different of the load's frequency spectrum [43]. Power splitting was here preferred to frequency splitting because it yielded better results.

SoC-targeted strategies

Two additional control laws were implemented to maintain the battery state of charge (SoC) around specific targets. The first one aims to keep SoC_{bat} close to 0.6 (denoted SoC06), representing a balanced state that allows both charging and discharging flexibility. The second seeks to maintain SoC_{bat} close to 1 (SoC1), favoring stored energy availability for the consumers but potentially increasing degradation due to sustained high SoC operation. These strategy philosophies are also very common approaches in hybrid electric systems [44][45].

Other rule-based strategies

More elaborate rule-based strategies were also implemented to balance dynamic responsiveness and computational simplicity. The first strategy (RB1) defines power sharing as a function of the battery SoC through piecewise rules. When the SoC is within the intermediate range ($0.2 < \text{SoC}_{\text{bat}} < 0.6$), the reference power demand $\hat{P}_{\text{tot,plan}}$ is split linearly between the battery and the hydrogen systems. The battery provides the full reference power when the SoC reaches 0.6, and none when it reaches 0.2. Outside this range, the battery operates alone when $\text{SoC}_{\text{bat}} > 0.6$ for discharging or when $\text{SoC}_{\text{bat}} < 0.2$ for charging, while the fuel cell or electrolyzer takes over as needed. This rule thus ensures smooth transitions between the different modes while preventing extreme SoC values. The second strategy (denoted RB2) simplifies the control logic by defining two main operating modes. When the total reference power $\hat{P}_{\text{tot,plan}}$ is positive (load demand exceeds generation), a fixed power offset $P_{\text{FC-offset}}$ is allocated to the fuel cell, and the battery provides the remainder. Similarly, when $\hat{P}_{\text{tot,plan}}$ is negative (excess production), the

electrolyzer absorbs a fixed power offset $P_{\text{ELY-offset}}$ of the total reference power, while the battery accounts for the rest. These parameters are tuned according to the best indicator values. This allows the hydrogen systems to operate at more constant powers, therefore extending their lifetime. These two rule-based strategies are graphically explained in Fig. 6.

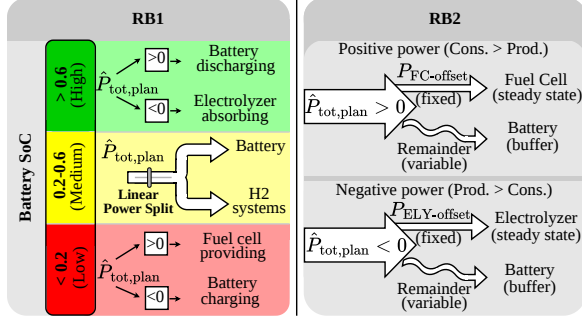


Figure 6: Visual explanation of the RB1 & RB2 strategies

All of the nine EMS structures are summarized in Table 3:

EMS Type	Denom.	Remarks
Power split	(X-Y)	$X, Y \in \{0, 25, 50, 75, 100\}$ [%], (battery-hydrogen shares).
SoC-target	SoCX	$X \in \{0.6, 1\}$: target SoC for batteries.
Others	RBX	$X \in \{1, 2\}$: logic from Fig. 6.

3. Results

3.1. EMS comparison in the Pareto plane

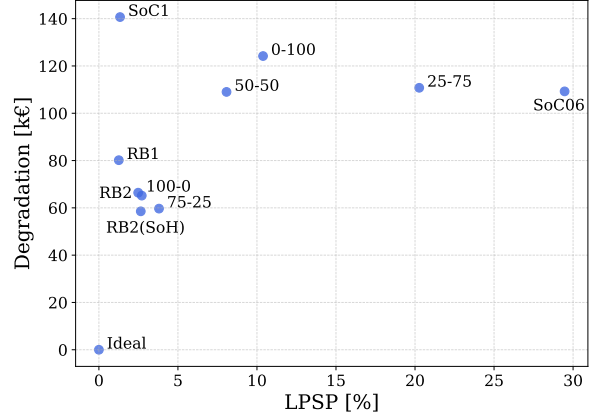


Figure 7: EMS visualisation in the Pareto plane for a 25-year simulation

Fig. 7 illustrates the positioning of the various EMS within the two-dimensional performance space defined by the service quality and sustainability indicators. Power-splitting strategies already seem relevant, but being as expected highly dependent on the share of power supplied by the battery. It has been observed that greater battery usage leads to a significant reduction (about 46.5% compared to the 0-100 EMS) in degradation costs, since hydrogen systems are especially prone to it. There is also an improvement in service quality (about 76.1%). The SoC-targeted strategies offer more extreme results. They can be both the best and the worst strategy considering LPSP, depending on the SoC target, but it always comes with the cost of high degradation. This is due to the fact that the hydrogen systems are dedicated to supporting the battery. The other rule-based EMS are also relevant, and among all tested strategies the rule-based RB2 looks to be providing the most balanced trade-off between service continuity and component longevity, making it the most promising baseline for further adaptive extensions presented in the next section.

3.2. Integration of component aging knowledge into the EMS

In this section, RB2 serves as a basis for integrating knowledge related to component aging into the control logic. The central idea of this enhancement is to dynamically adapt the EMS parameters according to the measured SoH of the components. In particular, the power of the PEMFC and PEMWE are adjusted as a function of their estimated SoH. This reduces the absolute value:

$$\begin{cases} \tilde{P}_{FC} = P_{FC\text{-offset}} \cdot \text{SoH}_{FC} & \text{if } \hat{P}_{\text{Load}} > \hat{P}_{PV} \\ \tilde{P}_{ELY} = P_{ELY\text{-offset}} \cdot \text{SoH}_{ELY} & \text{if } \hat{P}_{\text{Load}} < \hat{P}_{PV} \end{cases} \quad (14)$$

linking the reduction of delivered power to the reduction of maximum deliverable power of the components. The offset powers here are the target powers when the components are new, and as they degrade their ability to deliver the higher powers diminish. Consequently, the EMS progressively reduces the power assigned to them. This adaptation naturally maintains the operation of the hydrogen systems within lower power ranges, therefore mitigating further degradation. The evolution of the health-conscious RB2 in the Pareto plane is represented in Fig. 7 under the RB2(SoH) point. It can be observed that the degradation cost is markedly reduced (about -10%), at essentially constant service quality, the LPSP being even marginally lowered (by about 2%). This approach demonstrates how aging-informed control can enhance the durability of isolated hybrid microgrids at no expense for the short-term service. The evolution of the different power curves over different timescales can

be seen in Fig. 8, as well as the different aging quantities shown in Fig. 9.

3.3. Discussion

From the several results gathered in Fig. 7, it can be observed that for a timespan of twenty-five years, the best equivalent costs associated to the degradation of the components range from about 58 k€ to almost 141 k€, which underlines that the EMS choice is crucial for the operation of the system. These equivalent costs are linked to the fact that the components have to be replaced, this being at a rate of around once every 4 years for the battery, every 12 to 13 years for the fuel cell and every 19 years for the electrolyzer (see Fig. 9). These results of course stem from the SoH threshold chosen as the end-of-life (EoL) criteria, and is also heavily dependent on the sizing of the system.

This financial cost is associated with LPSP ranging from around 1% to around 29% for the more extreme ones. This can be directly read as the power supply on the side of the consumers being unavailable between 1% and 29% of the time, and is also naturally sizing-dependent. From the degradation charts, it can be observed that the start-stop events are by far the most significant factor contributing to degradation, accounting for about 96% of the PEMFC degradation and 55% of the PEMWE one, the remainder of the electrolyzer aging being dominated by the irreversible high-power contribution (about 39%). Tests have also been carried out to see whether keeping the hydrogen components in idling mode at low

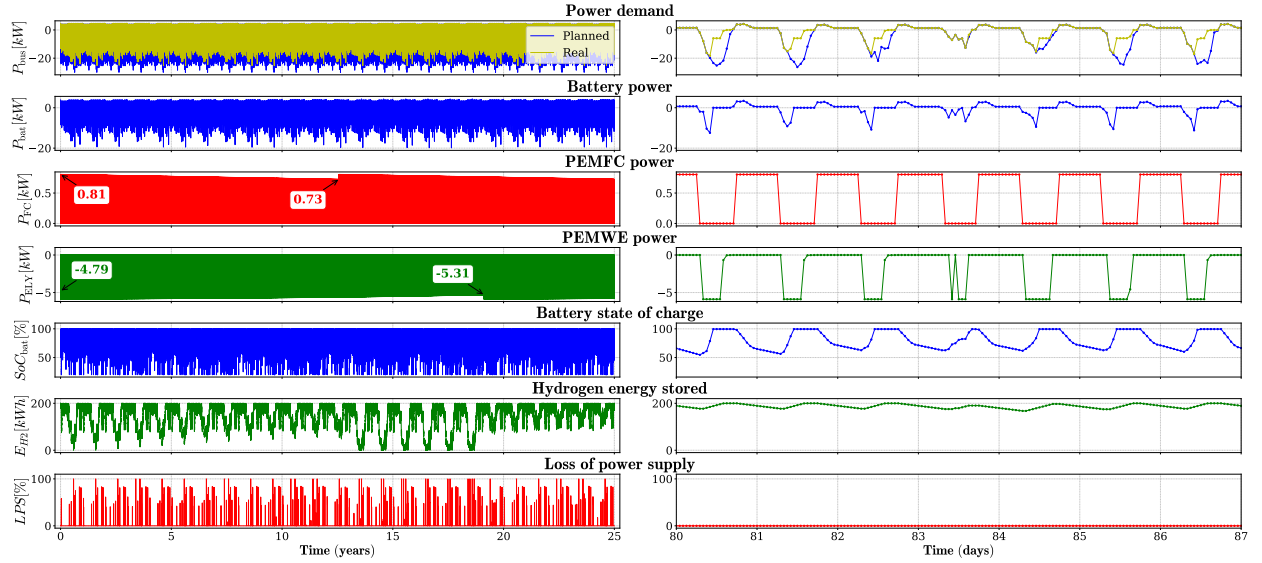


Figure 8: Power profiles over a 25-year simulation (left) and one week zoom (right), RB2(SoH) EMS

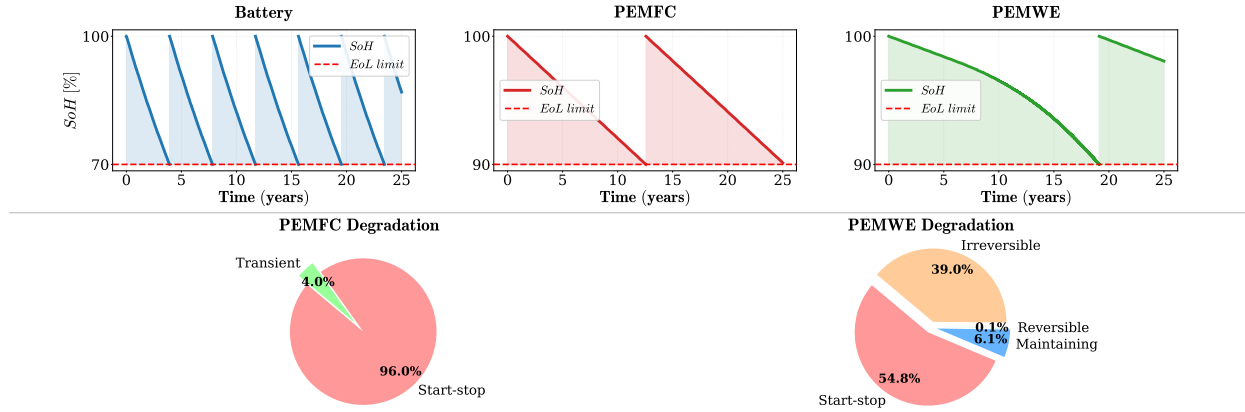


Figure 9: Evolution of the aging quantities, namely SoH and degradation types of the hydrogen components

demand, rather than shutting them down completely, could reduce damage. For both the PEMFC and the PEMWE, the results show the opposite: remaining in idling degrades the components more than performing a start-stop cycle.

It is to be noted that this work is mainly informative, in order to give the end user the means to make an informed choice. Some EMS are not strictly dominated in the Pareto sense, and therefore the choice for the most suited one may not be obvious. It will then be up to the local populations or the operator of the facility to choose whether the priority will be put on the financial cost associated to the long-term degradation or on the short-term convenience associated with a more trustworthy power supply. The results presented in this study are also to be tempered by the observed limitations inherent to this work. First, the results shown here are strongly associated with the power profiles that were used, those being given by actual data reflecting the consumption habits of local populations. For a deeper understanding of the different EMS performances, and increase the generality of these results, it would be of the utmost interest to cross-reference them with other results deduced from other power profiles. These profiles being thoughtfully constructed to reflect the specific reality of the case study, these EMS results remain relevant for the inhabitants of this microgrid in La Réunion.

Then, with a strong link to the power profiles used, the present analysis was performed at iso-sizing, with a simplified sizing of the components (PV array, battery, fuel cell, and electrolyzer). Although this approach enables clear comparison between control strategies, it does not necessarily reflect a physically optimized or economically realistic configuration. Revisiting the obtained results under a more detailed sizing process

would provide greater insight into the practical feasibility and scalability of the proposed EMS. This sizing aspect is also being studied inside the ANR-22-CE05-0026 project, of which this work is a part.

Also, the simulations were conducted with an hourly sampling period. This choice was motivated by the need to perform long-term simulations over several years while maintaining acceptable computational cost. However, such a coarse temporal resolution inevitably neglects short-term power fluctuations, which can have a non-negligible impact on the degradation dynamics of the electrochemical components. Consequently, the degradation indicator derived here may underestimate the true impact of these high-frequency variations.

4. Conclusion and perspectives

This study presented a methodological framework for the development and evaluation of an energy management strategy dedicated to an isolated hybrid microgrid composed of photovoltaic generation, a lithium-ion battery, and a hydrogen subsystem including a PEM fuel cell and electrolyzer. The proposed approach aimed to jointly address two key objectives that are often considered separately in the literature: ensuring short-term service continuity to end-users and preserving long-term component health to enhance system sustainability.

A set of various EMS were designed and compared within a single simulation environment. The assessment relied on two complementary indicators: LPSP, representing service unavailability, and a degradation cost reflecting cumulative aging effects on storage components. The comparative analysis demonstrated that a rule-based strategy that assigns more constant power

profiles to hydrogen components achieved the best compromise between short-term performance and long-term durability. Building on this, an adaptive version of this EMS was proposed, in which control parameters were dynamically updated according to the estimated state of health of the components. This enhancement enabled a significant reduction in replacement-related costs, without any penalty on power availability.

The results underline the benefit of integrating degradation knowledge directly into the energy management process. By embedding physical health information into the control logic, the EMS becomes capable of self-adjusting its operation to evolving system conditions, thereby maintaining performance over time while reducing wear on electrochemical devices. This also paves the way for predictive maintenance of the system. These results aim to be used by the local users or the microgrid operator who can thus choose which dimension to prioritize and make an informed choice.

Nevertheless, several limitations must be acknowledged. The results are inherently tied to the specific power profiles and the sizing used in this study. Extending the analysis to additional or synthetic profiles would improve the generality of the conclusions. Furthermore, the proposed framework focuses primarily on short-term and long-term objectives, without explicitly addressing intermediate timescales.

Future work will focus on extending this framework in several directions. First, integrating forecasting modules would allow the EMS to predict changes in PV production and user consumption, preventing degradation more effectively. Second, combining EMS with optimized microgrid sizing would strengthen the results and applicability. Lastly, incorporating a mid-term robustness metrics, for temporary failures, would contribute to a multi-scale approach uniting performance, reliability and durability.

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